House Price Prediction: Project Summary

Ritvik Gulati
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1. Objective & Problem Statement

The objective of this project was to develop a machine learning regression model capable of estimating a property's fair market value based on its physical characteristics and location. Additionally, the project aimed to identify the primary features that drive real estate prices to provide actionable business insights.

2. Data Preparation & Methodology

The analysis utilized a real-world housing dataset comprising 545 property records. The target variable was price, evaluated against 12 distinct property features.

- **Data Quality:** The dataset was highly complete, containing zero missing values and zero duplicate rows.
- **Feature Engineering:** Categorical variables (such as yes/no amenities and furnishing status) were transformed into numeric formats using one-hot encoding. All 12 original features were retained.
- **Modeling:** The data was split 80/20 into training and testing sets. Two predictive models were evaluated: Linear Regression and Random Forest Regressor.

3. Model Performance

Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R^2 Score
Linear Regression	₹ 970,043	₹ 1,324,507	0.6529
Random Forest	₹ 1,021,546	₹ 1,400,566	0.6119

Accuracy Assessment: The Linear Regression model explains approximately 65% of the variation in house prices. On average, predictions are off by roughly ₹9.7 lakh. Given a typical house price of ₹43 lakh in this market, this represents a variance of about 21–22%. This is highly useful for generating quick baseline estimates or flagging mispriced listings, though it is not a substitute for formal appraisals.

4. Key Findings & Insights

Primary Value Drivers

Based on correlation heatmaps and Random Forest feature importance tracking, area (square footage) and the number of bathrooms are the most dominant predictors of a house's price. Amenities like air conditioning, the number of stories, and parking capacity follow closely as strong secondary drivers.

Model Dynamics

Surprisingly, the simpler Linear Regression model outperformed the Random Forest Regressor. While Random Forest is generally considered a more robust model, the relatively small sample size of 545 houses likely caused the tree ensemble to overfit the training data. Furthermore, property prices in this specific dataset scale fairly linearly alongside the core features (area and bathrooms).

5. Business Recommendation

Stage and Furnish Properties Before Listing: The analysis highlighted that an "unfurnished" status is one of the few variables that tangibly drags a property's price down. Therefore, lightly staging or furnishing a property before taking listing photos or showing it to prospective buyers is a low-cost, high-impact strategy to elevate perceived market value and command a higher final sale price.